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↑ Personal Data:

- Surname: Jamali
- Name: Ramin
- Date and Place of Birth: November 21, 1995; Tehran, Iran
- Nationality: Iranian
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↑ Summary:

- I have a Ph.D. in Optics and Photonics from the Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran. My research expertise lies in advanced imaging and 3D microscopy and detection techniques, including speckle tweezers for manipulating micro-particles with varying refractive indices, polarization speckle field control, and optical trapping applications for studying colloidal forces and bacterial dynamics. I have hands-on experience in optical manipulation, structured light generation, force measurements in microfluidics systems, as well as the culture and control of microorganisms. These skills provide a strong foundation for applying state-of-the-art optical and microscopy techniques to investigate mechanobiological interactions in complex systems.

↑ Research Interests:

- 3D Imaging and Microscopies; Speckle Tweezers; Dynamic Speckle Pattern Analysis; Microfluidics, Depletion Forces in Colloidal Systems; Optical Manipulation.

↑ Education:

- Ph.D. in Optics and Photonics, 2021-2025, Institute for Advanced Studies in Basic sciences (IASBS), Zanjan, Iran
Supervisor: Dr. Ali-Reza Moradi

Thesis title: Polarization Speckle Tweezers and Speckle Trapping at Fluid-Fluid Interfaces

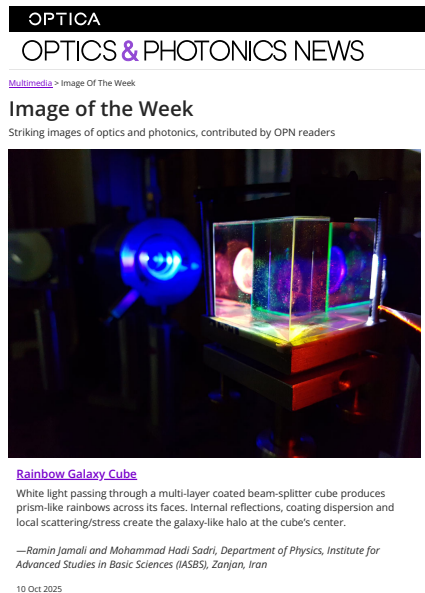
- M.Sc. in Optics, 2018-2021, Institute for Advanced Studies in Basic sciences (IASBS), Zanjan, Iran
Supervisors: Dr. Ali-Reza Moradi and Dr. Jalal Sarabadani

Thesis title: Investigation and Controlling of Colloidal Mixture by Speckle Tweezers

- B.Sc. in Physics, 2014-2018, University of Zanjan, Zanjan, Iran

↑ Awards and Honors:

- Selected as the October 2025 “Image of the Week” in the prestigious journal Optics and Photonics News (OPN), published by Optica (USA).
- Selected Ph.D. student of Elites Foundation (individual special project in Zanjan, Iran elites foundation), 2025
- The best annual Ph.D. student-researcher award in Zanjan province, 2024



- The best annual Ph.D. student-researcher award in Zanjan province, 2023
- Selected Ph.D. student of Elites Foundation (group special project in Zanjan, Iran elites foundation), 2023
- Selected Ph.D. student of Elites Foundation (individual special project in Zanjan, Iran elites foundation), 2023
- Selected Ph.D. student of Elites Foundation (individual special project in Zanjan, Iran elites foundation), 2022
- Top student (1st rank, GPA:19.51/20), Ph.D., Institute for advanced Studies in Basic Sciences, Zanjan, Iran, 2019
- Top student (1st rank, GPA:18.33/20), M.Sc., Institute for advanced Studies in Basic Sciences, Zanjan, Iran, 2019
- Top student (1st rank, GPA:17.87/20), B.Sc., University of Zanjan, Zanjan, Iran, 2018
- Top student (1st rank, GPA:17.52/20), High School, Allameh Amini, Zanjan, Iran, 2014
- Top student (1st rank, GPA:19.90/20), Guidance School, Kharazmi, Zanjan, Iran, 2010
- Top student (1st rank, GPA:20/20), Elementary School, Ehsan, Zanjan, Iran, 2007

↑ Computer and Programming Skills:

- Scientific Softwares: Matlab, Latex, Comsol Multiphysics, LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator)

- Programming Languages: C++ and Python
- Graphic Softwares: 3Ds Max, Google SketchUp, Adobe Photoshop, Adobe InDesign, ImageJ
- Electronic Softwares: Pspice, ProfiCAD
- Operating Systems : Linux (Ubuntu), Microsoft Office

↑ Activities:

- Teaching Assistant of Optics and Photonics I, January 2024-2025
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Modern Physics Laboratory, January 2024-2025
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Optics and Photonics I, January 2023-2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Optics and Photonics II, January 2023-2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of electromagnetism, January 2023-2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Electronics laboratory, January 2023-2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Electronics, January 2023-2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Modern Physics Laboratory, January 2023-2024
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Optics and Photonics, January 2021-2022
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Modern Physics Laboratory, April 2022
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Electronics, January 2022
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Electronics Laboratory, January 2018-2022
Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
- Teaching Assistant of Electronics Laboratory, January 2015-2018
University of Zanjan, Zanjan, Iran

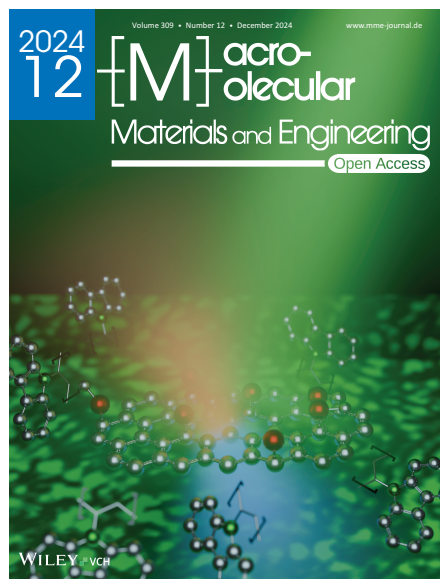
- Teaching Assistant of Electromagnetics, February 2017-2018
University of Zanjan, Zanjan, Iran
- Teaching Assistant of Electronics, October 2017-2018
University of Zanjan, Zanjan, Iran
- Teaching Assistant of Optics and Laser, October 2017-2018
University of Zanjan, Zanjan, Iran
- Teaching Assistant of Physics II (Electromagnetics), February 2016-2017
University of Zanjan, Zanjan, Iran

↑ Involved Research Projects:

1. Speckle Tweezers in Liquid-Liquid Interface
 - Collaboration with Dr. ALi-Reza Moradi and Dr. Sabareesh K. P. Velu's group (Rathinam College of Arts and Science, India)
2. Controlling Collective Motion by Speckle Tweezers
 - Collaboration with Dr. ALi-Reza Moradi and Dr. Sabareesh K. P. Velu's group (Rathinam College of Arts and Science, India)
3. 3D Imaging Using Scanning Diffractometry with Speckle Pattern Analysis
 - Collaboration with Dr. ALi-Reza Moradi.
4. Revealing the structure deformation in multi-component lipids by dynamic speckle pattern analysis
 - Collaboration with Dr. Ahmad Darudi (University of Zanjan, Iran).
5. The Role of Magnetic Field Effect on Morphology and Growth Rate of Myelin Figures
 - Collaboration with Dr. ALi-Reza Moradi.
6. Metamaterial Assisted Illumination Nanoscopy via Random Super-Resolution Speckles
 - Collaboration with Dr. ALi-Reza Moradi.
7. Phase Separation Detection of Binary Mixture of by Speckle Pattern Analysis
 - Collaboration with Dr. ALi-Reza Moradi.

↑ Publications-Journals:

16. R. Jamali, M. H. Sadri, and A. R. Moradi, “Remote characterization of aerogel foam concrete using dynamic speckle pattern analysis”, Under review in Cement and Concrete Research, 2025
15. R. Jamali, M. H. Sadri, and A. R. Moradi, “Speckle tweezers in action: manipulating run-and-tumble in bacteria”, Under review in Optics Letters, 2025
14. R. Jamali, S. K. P. Velu, and A. R. Moradi, “Speckle tweezers near water-oil and water-air interfaces”, Under review in Biomedical Optics Express, 2025
13. M. H. Sadri, R. Jamali, A. J. Khan, F. Rehman and A. R. Moradi, “Sorting of mesoporous silica derivatives by random optical fields”, Under review in iScience, 2025
12. M. H. Sadri, R. Jamali, and A. R. Moradi, “Speckle fields for controlling and analysis of microscopic collective motion”, Optics Express, Vol. 33, Issue 11, pp. 22864-22882, 2025
11. M. Sajjadi, R. Jamali, T. Kiyani, Z. Mohamadnia and A. R. Moradi, “Characterization of Schiff base self-healing hydrogels by dynamic speckle pattern analysis”, Scientific Reports, 14 (1), 27950, 2024
10. R. Jamali, M. Sajjadi, B. Taherkhani, D. Abbaszadeh and A. R. Moradi, “Front Cover: Speckle pattern analysis of PVK graphene memristor”, Macromolecular Materials and Engineering, December 2024



9. R. Jamali, M. Sajjadi, B. Taherkhani, D. Abbaszadeh and A. R. Moradi, “Speckle pattern analysis of PVK graphene memristor”, Macromolecular Materials and Engineering, 6:2400213, 2024

8. R. Jamali, V. Farzam Rad, M. Razaghi, Z. Mohamadnia, M. Khorasani and A. R. Moradi, "Digital holographic microscopy of spiropyran-based dynamic materials", *Journal of Microscopy*, 292(2), 2023
7. R. Jamali, A. Babaei-Ghazvini, E. Nazari, M. Panahi, I. Shahabi-Ghahfarrokhi, and A. R. Moradi, "Surface characterization of biodegradable nanocomposites by dynamic speckle analysis", *Applied Surface Science Advances*, 16 (100429), 2023
6. O. Pedram, R. Jamali, V. Farzam Rad, R. Khamedi, E. Poursaeidi and A. R. Moradi, "Evaluation of pitting corrosion by dynamic speckle pattern analysis", *Sci. Rep.* 26;13(1):8549, 2023
5. M. Panahi, V. Farzam Rad, Sh. Sasan, R. Jamali, A. R. Moradi and A. Darudi, "Detection of intralayer alignment in multi-component lipids by dynamic speckle pattern analysis", *Journal of Biophotonics*, <https://doi.org/10.1002/jbio.202200034>, 2022
4. R. Jamali, F. Nazari, A. Ghaffari, S. K. P. Velu, and A. R. Moradi, "Speckle tweezers for manipulation of high and low refractive index micro-particles and nano-particle loaded vesicles", *Nanophotonics*, 10 ,11, pp. 2915-2928 ,2021
3. M. Panahi, R. Jamali, V. Farzam Rad, M. Khorasani, A. Darudi, and A. R. Moradi, "3D monitoring of the surface slippage effect on micro-particle sedimentation by digital holographic microscopy", *Sci. Rep.* 11, 12916, 2021
2. V. Farzam Rad, A. Babaei-Ghazvini, R. Jamali, I. Shahabi-Ghahfarrokhi, and A. R. Moradi, "Digital holographic microscopy for real-time investigation of 3D microstructural dynamics of starch-kefir based nanocomposite", *Appl. Opt.* 60(16), pp. 4706-4715, 2021
1. V. Farzam Rad, M. Panahi, R. Jamali, A. Darudi, and A. R. Moradi, "Non-invasive in situ monitoring of bone scaffold activity by speckle pattern analysis", *Biomed. Opt. Express*, 11(11), pp. 6324-6336, 2020

↑ Publications-Proceedings-International:

6. V. Abbasian, R. Jamali, V. Farzam Rad, A. R. Moradi, and A. Darafsheh “Dynamic speckle pattern analysis of pitting corrosion”, Photonic Instrumentation Engineering XI 12893, 238-241, March 2024
5. M. H. Sadri, R. Jamali, F. Rehman, A. R. Moradi, “Mesoporous material sorting in random energy landscapes”, 11th International Symposium on Optics and its Applications OPTICS11, 89, July 2023
4. R. Jamali, B. Olamaei, M. Soltanloo, S. E. H. Yeganeh, V. Farzam Rad, J. Esmkhani, and A. R. Moradi, “ Measurement of corrosion in power plant turbine rotors by digital holographic microscopy”, Imaging Congress, hybrid , 14 - 17 August 2023
3. E. Nasimdoust R. Jamali, F. Amarloo, S. K. P. Velu, and A. R. Moradi, “ Controlling Spatial Distribution of Speckle Fields for Multi-Manipulation of Colloids”, Optical Trapping and Optical Micromanipulation XIX, 21 - 24 August 2022
2. R. Jamali, F. Nazari, A. Ghaffari, S. K. P. Velu, and A. R. Moradi, “ Simultaneous Manipulation of Low and High Refractive Index Micro-particles by Speckle Tweezers”, PSL Summer School on Soft and Living matter, July 2021
1. R. Jamali, F. Nazari, A. Ghaffari, S. K. P. Velu, and A. R. Moradi, “Collective Manipulation of Low and High Refractive Index Micro-particles by Speckle Tweezers”, International Virtual Conference on Novel nanomaterials for innovative reaserch ICNNIR, April 2021

↑ Publications-Proceedings-National:

7. M. H. Sadri, R. Jamali, F. Rehamn, and A. R. Moradi, “Mesoporous silica sorting by speckle tweezers”, Proc. 29th Iranian Conference on Optics and Photonics and 15th Iranian Conference on Engineering Photonic (ICOP29), February 2023
6. R. Jamali, and A. R. Moradi, “Experimental Study of Liposome Aggregation”, Proc. 26th Annual IASBS Meeting on Condensed Matter Physics, July 2021
5. R. Jamali, T. Ohadi, and A. R. Moradi, “Investigating the Time Evolution of Color Distribution in a Variety of Surfaces”, Proc. 26th Annual IASBS Meeting on Condensed Matter Physics, July 2021
4. M. Karimi, R. Jamali, A. Bayat, Y. Abedini, Y. Ghaniun, “Building One Channel Sun-Photometer Using Light Emitting Diode”, Proc. 26th Iranian Conference on Optics and Photonics and 12th Iranian Conference on Engineering Photonic (ICOP19), February 2020

3. R. Jamali and S. Mollaei, "Time Evolution of Size Distribution for Oil Droplets on Water", Proc. 25th Annual IASBS Meeting on Condensed Matter Physics and School on Topological Phases of Matter, June 2019
2. A. Bayat, R. Jamali and S. Mollaei, "Calculation of Rotational Inertia of Gyroscope and Measurement of Force Vector by Precession," Proc. 17th Zanzan Education Research Journal, July 2018
1. S. Mollaei, R. Jamali and A. H. Daroneh, "Investigation of Self- Assembly of Aluminium Flakes on Water Surface", Proc. 24th Annual IASBS Meeting on Condensed Matter Physics and School on Topological Phases of Matter, June 2018

↑ Talks:

- “Speckle Tweezers”, Physics Group Seminars, University of Zanjan, Zanjan, Iran, November 2025
- “Speckle Tweezers at Fluid-Fluid Interfaces”, Optics Group Seminars, IASBS, Zanjan, Iran, Feb 2024
- “Speckle Tweezers”, NanoQIQO School on Optics and Photonics (SOP) 2023, Yerevan, Armenia, May 2023
- “Investigation and Controlling of Colloidal Mixture by Speckle Tweezers”, Optics Group Seminars, IASBS, Zanjan, Iran, June 2021
- “Designing and Construction of a LED Sun Photometer”, University of Zanjan, Zanjan, Iran, June 2017
- “Calculation of Rotational Inertia of Gyroscope and Measurement of Force Vector by Precession”, University of Zanjan, Zanjan, Iran, December 2016

↑ Schools and Conferences Attended:

13. NanoQIQO School on Optics and Photonics (SOP), Yerevan, Armenia May 15-20, 2023
12. PSL Summer School on Soft and Living matter, Paris, France July 5-9th, 2021
11. 26th Annual IASBS Meeting on Condensed Matter Physics, IASBS, Zanjan, Iran, July 7-9th, 2021
10. RSC Desktop Seminar Lecturship with Soft Matter, RSC webinars, April 29th, 2021
9. International Virtual Conferance on “Novel Nanomaterials for Innovative Reaserch ICNNIR, 29th & 30th April, 2021
8. One-day Online Workshop on Novel Features and Applications of Optical Manipulation, IPM, Tehran, Iran, September 8th, 2020
7. 26th Iranian Conference on Optics and Photonics and 12th Iranian Conference on Engineering Photonic, Kharazmi University (ICOP19), February, 2020
6. 25th Winter School of Physics, IASBS, Zanjan, Iran, February, 2020
5. 25th Annual IASBS Meeting on Condensed Matter Physics, IASBS, Zanjan, Iran, June 13-14th, 2019
4. 24th Annual IASBS Meeting on Condensed Matter Physics, IASBS, Zanjan, Iran, June 13-14th, 2018
3. Workshop on Non-equilibrium Soft Matter, IPM, Tehran, Iran, April 11th, 2019

2. Center for Research in Climate Change And Global Warming (CRCC), University of Zanjan, Zanjan, Iran, December 2015
1. The First Workshop on The Regional Climate Model system (RegCM), December 2015

↑ Language Skills:

- Mother Tongue (Bilingual): Persian and Azeri
 - English Language
 - Understanding
 - Listening: C1
 - Reading: C1
 - Speaking
 - Spoken Interaction: B1
 - Spoken Production: B1
 - Writing
 - Writing Skills: B2
- Levels: A1/A2: Basic user - B1/B2: Independent user - C1/C2 Proficient user
- Good Communication Skills, Active Listening, Clarifying and Summarising
 - Good Command of Quality Data Processing and Circuit Design

↑ Further Activities:

- Persian Calligraphy
- Watching Movies & Series
- Graphic Design with 3Ds Max

↑ Referees:

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<http://www.znu.ac.ir/members/bayat-al>